

SKILL-5

Roll No. : 2520030437

Name : Bandla Venkata Rama Vinay

SEC-8A

SKILL-5:

Aim: Develop a C program to identify whether a command is enclosed in single or double quotation marks and demonstrate the different handling of literal content and variable expansion.

The program accepts a quoted command, preserves the entered content, detects the quotation type, and checks for the **\$HOME** variable when double quotes are used.

Procedure :

1. Create the C source file **Skill-5.c**.
2. Read the quoted command using **fgets()** and remove the trailing newline using **strcspn()**.
3. Check the first character of the input to identify single or double quotation marks.
4. For a single quoted string, report that literal content is preserved and variable expansion is ignored.
5. For a double quoted string, preserve spaces and search for **\$HOME** using **strstr()**.
6. If **\$HOME** is found, obtain its value using **getenv()** and display the expanded value.

Output :

```
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ ./Skill_5
Enter a quoted command: '$HOME hello world'

Input: '$HOME hello world'
Type: Single Quoted String
Literal content preserved.
Variable expansion ignored.
vinay@DESKTOP-0CVM4SE:/mnt/c/Testing/OSSP_Testing/skill$ |
```

The submitted execution enters the quoted command '**\$HOME hello world**'. The program identifies it as a **Single Quoted String**, reports that the literal content is preserved, and states that variable expansion is ignored.

Code:

```
Skill > c Skill-5.c
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  int main()
6  {
7      char input[200];
8
9      printf("Enter a quoted command: ");
10     fgets(input, sizeof(input), stdin);
11
12     input[strcspn(input, "\n")] = '\0';
13
14     printf("\nInput: %s\n", input);
15
16     if (input[0] == '\')
17     {
18         printf("Type: Single Quoted String\n");
19         printf("Literal content preserved.\n");
20         printf("Variable expansion ignored.\n");
21     }
22     else if (input[0] == '"')
23     {
24         printf("Type: Double Quoted String\n");
25         printf("Spaces preserved.\n");
26
27         char *pos = strstr(input, "$HOME");
28
29         if (pos != NULL)
30         {
31             printf("Variable found: $HOME\n");
32             printf("Expanded value: %s\n", getenv("HOME"));
33         }
34         else
35         {
36             printf("No variable found.\n");
37         }
38     }
39 }
```

Code (continued):

```
38     }
39     else
40     {
41         printf("No quotation detected.\n");
42     }
43
44     return 0;
45 }
```

Functions and Program Operations

Function / Operation	Purpose
fgets()	Reads the quoted command from standard input.
strcspn()	Finds the newline character so it can be removed.
strlen()	Checks whether the entered input is empty.
strstr()	Searches the input for the literal string "\$HOME".
getenv()	Obtains the value of the HOME environment variable.

Quotation Handling

When the first character is a single quote, the program classifies the input as a **Single Quoted String**, states that literal content is preserved, and ignores variable expansion.

When the first character is a double quote, the program classifies the input as a **Double Quoted String**, preserves spaces, searches for **\$HOME**, and displays its environment value when the variable is found.

Observed Execution

'\$HOME hello world' → Single Quoted String → Literal content preserved → Variable expansion ignored.

Observation :

- The program accepts a quoted command through standard input.
- The trailing newline from **fgets()** is removed before processing.
- The program distinguishes between single and double quoted input using the first character.
- Single quoted input preserves literal content and does not perform variable expansion.
- Double quoted input preserves spaces and checks for the **\$HOME** variable.
- The submitted execution correctly classifies '**\$HOME hello world**' as a Single Quoted String.

Result :

The C program successfully demonstrates quotation handling for shell-style command input. The submitted test correctly identifies the single quoted string, preserves the literal **\$HOME** text, and reports that variable expansion is ignored.

SKILL-5 Completed Successfully